

CUSTOMER SHOPPING BEHAVIOR

Business Analysis Report

Python (pandas) • PostgreSQL • Power BI

1. Project Overview

This project analyzes shopping behavior across 3,900 customer transactions to understand spending patterns, customer segments, product performance, and the impact of subscriptions and discounts. The analysis combines Python for data cleaning and exploration, PostgreSQL for structured business-question queries, and Power BI for an interactive summary dashboard. The objective is to translate the findings into concrete, actionable recommendations for revenue growth and customer retention.

2. Executive Summary

3,900	\$59.76	3.75 / 5	27%
Total Customers	Avg. Purchase Amount	Avg. Review Rating	Subscription Rate

The dataset skews heavily toward non-subscribers (73%), who nonetheless generate the majority of total revenue simply due to their larger numbers — subscribers and non-subscribers spend almost identically per person. The most loyal customers (5+ purchases) are overwhelmingly unsubscribed, pointing to a clear, low-risk opportunity to convert existing loyal behavior into recurring subscription revenue. Clothing and Young Adult customers are the strongest performing category and segment respectively, and several products (Hats, Sneakers, Coats) rely heavily on discounting to sell — a margin risk worth monitoring.

3. Methodology

3.1 Data Preparation (Python)

- **Data Loading & Exploration:** Loaded and inspected 3,900 rows × 18 columns using pandas (df.info(), df.describe()).
- **Missing Data Handling:** 37 missing Review Rating values were imputed using the median rating for each product category.
- **Standardization:** Columns renamed to snake_case for consistency and downstream SQL compatibility.
- **Feature Engineering:** Created age_group (binned ages) and purchase_frequency_days; dropped the redundant promo_code_used column after confirming overlap with discount_applied.
- **Database Integration:** Loaded the cleaned DataFrame into PostgreSQL via SQLAlchemy for structured SQL analysis.

3.2 Business Analysis (PostgreSQL)

Ten SQL queries were run against the cleaned dataset to answer specific business questions, summarized in Section 4.

4. Key Findings

4.1 Revenue by Gender

Gender	Total Revenue (\$)
Female	75,191
Male	157,890

Insight: Male customers generate more than double the revenue of female customers ($\approx 68\%$ of total revenue). This is a strong signal worth validating against category preferences — marketing spend and inventory planning should reflect this skew rather than assuming an even split.

4.2 Subscribers vs. Non-Subscribers

Status	Customers	Avg. Spend (\$)	Total Revenue (\$)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Insight: Average spend per customer is virtually identical between subscribers and non-subscribers (\$59.49 vs \$59.87). Subscription status is therefore not currently associated with higher spending — it functions only as a headcount effect. This means the subscription program is not yet doing its job of driving incremental spend, and represents unrealized value rather than a working loyalty lever.

4.3 Repeat Buyers & Subscription Status

Subscription Status	Repeat Buyers (5+ purchases)
No	2,518
Yes	958

Insight: Nearly 3 out of 4 repeat buyers are NOT subscribed. These are customers who already behave loyally without any subscription incentive — they are the easiest, lowest-risk group to convert into paying subscribers, since the habit already exists.

4.4 Customer Segmentation

Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

Insight: 80% of the customer base is already classified as Loyal, which is a strong retention signal for the business overall — but only 27% of that same base is subscribed (Section 4.2). The gap between behavioral loyalty and subscription enrollment is the single biggest opportunity in this dataset.

4.5 Revenue by Age Group

Age Group	Total Revenue (\$)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Insight: Revenue is fairly evenly distributed across age groups, with Young Adults leading only modestly ($\approx$ \$6,400 ahead of the lowest group, Senior). This suggests the product appeals broadly across generations rather than skewing to one segment — age-targeted campaigns should be tested carefully, since the uplift potential from age targeting alone appears limited.

4.6 Shipping Type & Spend

Shipping Type	Avg. Purchase Amount (\$)
Standard	58.46
Express	60.48

Insight: Customers who choose Express shipping spend about 3.5% more on average than Standard shipping customers. While modest, this is consistent with willingness-to-pay behavior and supports testing express shipping as an upsell at checkout.

4.7 Top-Rated Products

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Ratings across the top 5 products are tightly clustered (3.78–3.86), indicating no single standout product — but also no clear underperformer among top sellers. These products are safe to feature in marketing since satisfaction is consistently above average.

4.8 Discount-Dependent Products

Product	% of Purchases Discounted
Hat	50.0%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%

Product	% of Purchases Discounted
Pants	47.4%

Insight: Roughly half of all purchases of these five products involve a discount. Notably, Hats appear on both this list and the top-rated list — well-liked but heavily discount-reliant — which is a useful combination to flag for a controlled price test, since demand may hold up even at a smaller discount.

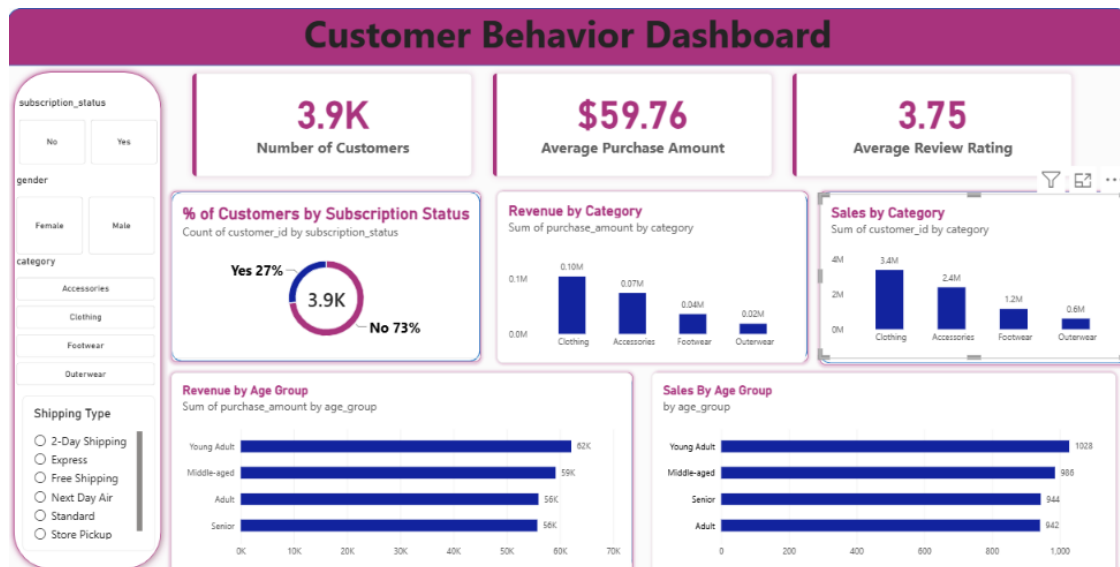
4.9 Top Products per Category

Category	Top Product	Orders
Accessories	Jewelry	171
Clothing	Blouse / Pants (tie)	171
Footwear	Sandals	160
Outerwear	Jacket	163

Insight: Clothing and Accessories categories show the highest order counts for their top products, reinforcing Clothing's position (seen on the dashboard) as the leading category by both revenue and units sold — a strong candidate for continued inventory investment.

5. Interactive Dashboard (Power BI)

The findings above are consolidated into an interactive Power BI dashboard, allowing filtering by subscription status, gender, category, and shipping type.



6. Business Recommendations

- **Convert Loyal Behavior into Subscriptions:** Repeat buyers are already loyal in behavior but not enrolled — target the 2,518 unsubscribed repeat buyers directly with a low-friction subscription offer (e.g., first-month discount or free express shipping) rather than marketing subscriptions broadly.

- **Redesign Subscription Value:** Since subscribers and non-subscribers currently spend the same on average, add tangible perks (early access, subscriber-only discounts, free express shipping) so subscription status starts driving incremental spend rather than just recurring billing.
- **Audit Discount-Dependent Products:** Hats, Sneakers, and Coats sell well but rely on discounts roughly half the time. Run a controlled A/B price test at a lower discount depth for these items to check whether demand holds, protecting margin.
- **Investigate the Gender Revenue Gap:** Male customers drive roughly two-thirds of revenue — investigate underlying category/product preferences by gender to see whether this is a marketing-reach gap or a genuine demand difference, and adjust campaign spend accordingly.
- **Double Down on Clothing:** Clothing is the top-performing category on both revenue and units sold; prioritize it in campaigns and ensure inventory levels support continued demand, especially for Blouses and Pants.
- **Test a Shipping Upsell:** Given the ~3.5% average spend premium among Express shipping customers, test offering Express as a discounted upsell at checkout rather than a flat-rate option, to nudge more customers toward the higher-spend behavior.

7. Conclusion

The data points to one consistent theme: the business already has a large, loyal customer base, but its subscription program is not yet capturing or rewarding that loyalty. The highest-leverage next step is closing the gap between behavioral loyalty (80% of customers) and subscription enrollment (27%), since this segment requires no new customer acquisition — only a better-targeted offer to customers who are already buying repeatedly. Secondary opportunities in discount policy, gender-based targeting, and shipping upsells offer smaller but low-effort revenue gains that can be tested in parallel.